1 - Étude d'une fonction

Soient f, g et h les fonctions définies par $f(x) = e^{-x^2}$, $g(x) = -x^4 + 2x^2 + 1$ et $h(x) = \frac{x+1}{x-2}$.

x	$-\infty$		$+\infty$
$f(x)$		$+$	

x	$-\infty$		$-\sqrt{1+\sqrt{2}}$		$\sqrt{1+\sqrt{2}}$		$+\infty$
$x^4 - 2x^2 - 1$		$+$		$-$		$+$	
$g(x)$		$-$	0	$+$	0	$-$	

x	$-\infty$		-1		2		$+\infty$
$x+1$		$-$	0	$+$		$+$	
$x-2$		$-$		$-$	0	$+$	
$h(x)$		$+$	0	$-$	$\|\|$	$+$	

x	$-\infty$		0		$+\infty$
$f'(x)$		$+$	0	$-$	
$f(x)$	0	$\nearrow$	1	$\searrow$	0

x	$-\infty$		-1		0		1		$+\infty$
$g'(x)$		$+$	0	$-$	0	$+$	0	$-$	
$g(x)$	$-\infty$	$\nearrow$	2	$\searrow$	1	$\nearrow$	2	$\searrow$	$-\infty$

x	$-\infty$		2		$+\infty$
$h'(x)$		$-$	$\|\|$	$-$	
$h(x)$	1	$\searrow$	$-\infty \,\|\|\, +\infty$	$\searrow$	1

$f'(x) = -2xe^{-x^2}$

$g'(x) = 4x\left(1 - x^2\right)$

$h'(x) = -\dfrac{3}{(x-2)^2}$

$f''(x) = 2\left(2x^2 - 1\right)e^{-x^2}$

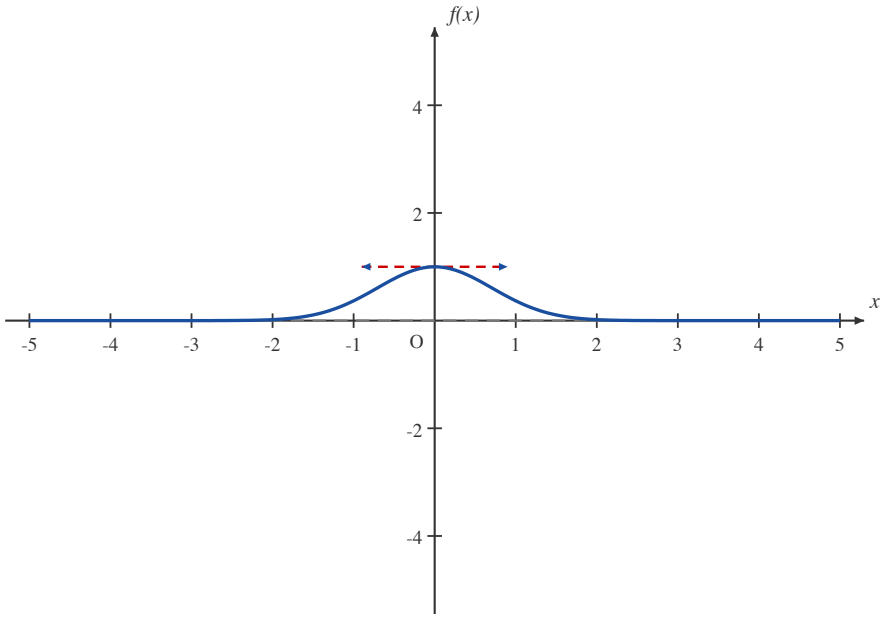
$g''(x) = 4 - 12x^2$

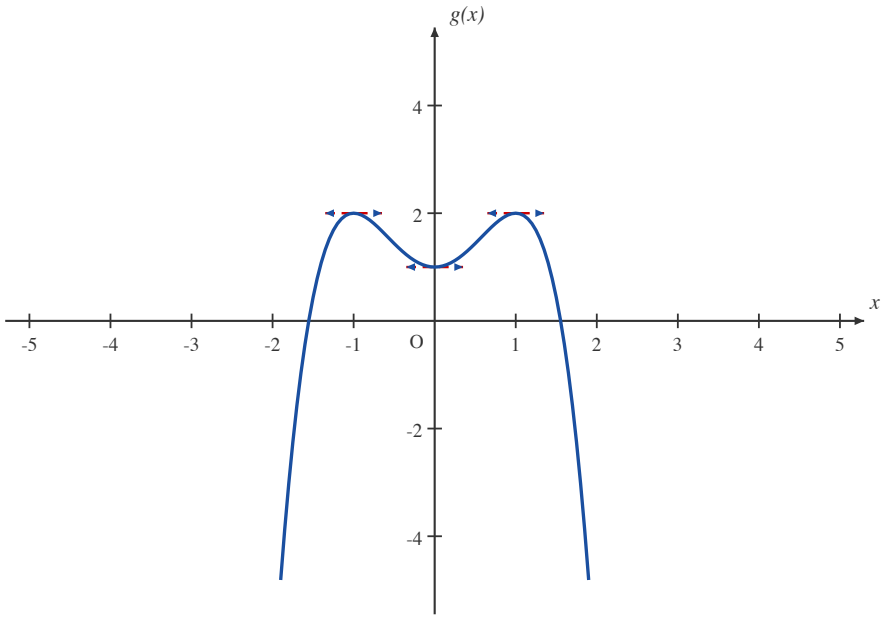
$h''(x) = \dfrac{6}{(x-2)^3}$

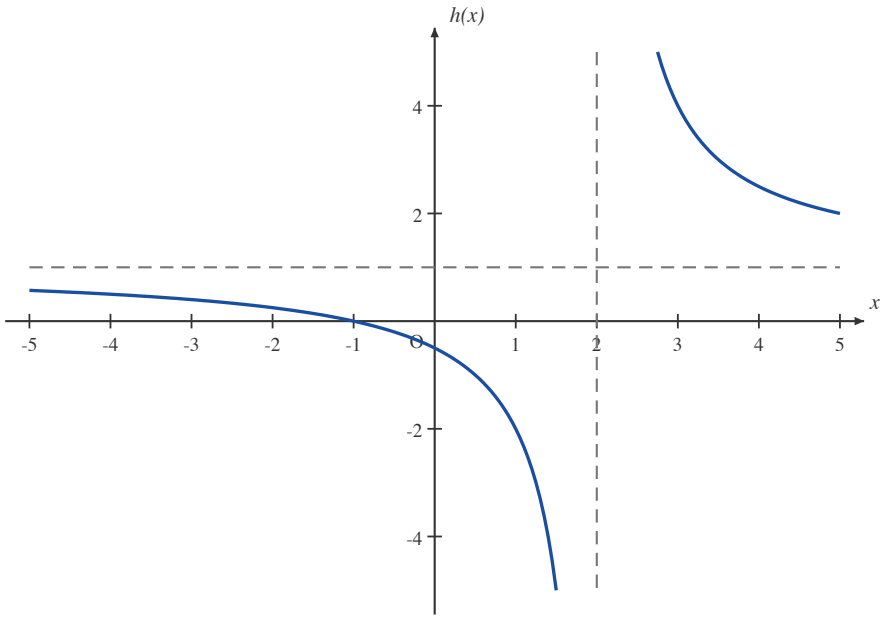
$f(x) = 0 \iff x \in \{\varnothing\}$

$g(x) = 0 \iff x \in \left\{-\sqrt{1+\sqrt{2}}, \sqrt{1+\sqrt{2}}\right\}$

$h(x) = 0 \iff x \in \{-1\}$


f(x)
x
O
-5
-4
-3
-2
-1
1
2
3
4
5
4
2
-2
-4


g(x)
x
O
-5
-4
-3
-2
-1
1
2
3
4
5
4
2
-2
-4


h(x)
x
O
-5
-4
-3
-2
-1
1
2
3
4
5
4
2
-2
-4